

Total No. of Questions : 8]

SEAT No. :

P814

[5870] - 1135

[Total No. of Pages : 2

**T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - II) (310253)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q 1 or Q 2, Q 3 or Q 4, Q 5 or Q 6, Q 7 or Q 8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Assume suitable data if necessary.

Q1) a) Explain Alpha - Beta Tree search and cutoff procedure in detail with example. [9]

b) What are the issues that need to be addressed for solving esp efficiently? Explain the solutions to them. [9]

OR

Q2) a) Explain in detail the concepts of back tracking and constraint propagation and solve the N-queen problem using these algorithms. [9]

b) Write a short note on Monte Carlo Tree search and list its limitations. [5]

c) Apply constraint satisfaction method to solve following Problem

SEND + MORE = MONEY. (TWO + TWO = FOUR, CROSS+ROADS= DANGER) [4]

Q3) a) List the inference rules used in propositional logic? Explain them in detail with suitable example. [9]

b) Explain syntax and semantics of First Order Logic in detail. [8]

OR

Q4) a) Detail the algorithm for deciding entailment in propositional logic. [8]

b) Explain knowledge representation structure and compare them. [9]

P.T.O.

- Q5)** a) Explain Forward and Backward chaining. What factors justify whether reasoning is to be done in forward or backward chaining. [9]
- b) What are the reasoning patterns in propositional logic? Explain them in detail. [9]

OR

- Q6)** a) Explain unification algorithm with an example. [8]
- b) Explain knowledge representation structures and compare them. [7]
- c) What do you mean by Ontology of situation calculus? [3]
- Q7)** a) Analyse various planning approaches in detail. [9]
- b) Discuss AI and its ethical concerns. Explain limitations of AI. [8]

OR

- Q8)** a) Explain the terms for time and schedule from perspective of temporal planning. [9]
- b) Write a detailed note on AI Architecture. [8]

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Total No. of Questions : 8]

SEAT No. :

PA-1451

[Total No. of Pages : 2

[5926]-67

T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - II) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data, if necessary.*

Q1) a) Explain Min Max and Alpha Beta pruning algorithm for adversarial search with example. **[9]**

b) Define and explain Constraints satisfaction problem. **[9]**

OR

Q2) a) Explain with example graph coloring problem. **[9]**

b) How AI technique is used to solve tic-tac-toe problem. **[9]**

Q3) a) Explain Wumpus world environment giving its PEAS description. **[9]**

b) Explain different inference rules in FOL with suitable example. **[8]**

OR

Q4) a) Write an propositional logic for the statement, **[10]**

i) "All birds fly"

ii) "Every man respect his parents"

b) Differentiate between propositional logic and First order logic. **[7]**

P.T.O.

- Q5)** a) Explain Forward chaining algorithm with the help of example. [9]
b) Write and explain the steps of knowledge engineering process. [9]

OR

- Q6)** a) Explain Backward chaining algorithm with the help of example [9]
b) Write a short note on : [9]
i) Resolution and
ii) Unification

- Q7)** a) Write a short note on planning agent, state goal and action representation. [6]
b) Explain different components of planning system. [6]
c) Explain the components of AI. [5]

OR

- Q8)** a) What are the types of planning? Explain in detail. [6]
b) Explain Classical Planning and its advantages with example. [6]
c) Write note on hierarchical task network planning. [5]



Total No. of Questions : 8]

SEAT No. :

P277

[Total No. of Pages : 2

[6003]:356

T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - II) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3, or Q.4, Q.5 or Q.6 Q.7, or Q.8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Assume suitable data if necessary.

- Q1)** a) List All problem solving strategies. What is backtracking, explain with n queen problem, with Branch and bound or Backtracking. [8]
- b) Explain Monte Carlo Tree Search with all steps and Demonstrate with one Example. [9]

OR

- Q2)** a) i) Explain limitations of game search algorithm, Differentiate between stochastic and partial games AND. [9]
- ii) Explain How use of alpha and beta cut-offs will improve performance of mini max algorithm? [9]
- b) Define is Constraint satisfaction problem, State the types of consistencies Solve the following Crypt Arithmetic Problem. [8]

SEND

+MORE

MONEY

- Q3)** a) What is an Agent Name any 5 agents around you Explain Knowledge based agent with Wumpus World. [9]
- List and explain in short the various steps of knowledge engineering process.
- b) Consider the following axioms: [9]
- If a triangle is isosceles, then its two sides AB and AC are equal,
If AB and AC are equal, then angle B and C are equal
ABC is an equilateral triangle,
Represent these facts in predicate' logic.
Explain Inference in Propositional Logic.

OR

P.T.O.

Q4) a) Write the following sentences in FOL (any 2) (using types of quantifiers). [9]

- i) Every number is either negative or has a square root .
- ii) Every connected and circuit-free graph is a tree .
- iii) Some people are either religious or pious
- iv) There is a barber who shaves all men in the town who do not shave themselves

b) What is Resolution? Solve the following statement by using resolution algorithm. Draw suitable resolution graph. [9]

- i) Rajesh like all kind of food.
- ii) Apple and vegetables are food.
- iii) Anything anyone eats and is not killed is food.
- iv) Ajay eats peanuts and still alive.

Prove that Rajesh like bananas. .

Q5) a) Explain Forward Chaining and Backward Chaining. With its Properties, with one. example. [9]

b) Explain Unification Algorithm in FOL. Solve stepwise with proper comments if $p(x, g(x))$ is equal to or not equal to $f(\text{prime}, f(\text{prime}))$ [8]

OR

Q6) a) Explain FOL inference for following Quantifiers. [8]

- i) Universal Generalization.
- ii) Universal Instantiation.
- iii) Existential Instantiation.
- iv) Existential introduction

b) What is Ontological Engineering ,in details with its categories object and Model. [9]

Q7) a) Explain with an example State Space Planning. [5]

b) Explain with example, how planning is different from problem solving. [5]

c) Explain AI components and AI architecture. [8]

OR

Q8) a) Explain Planning in non deterministic domain. [5]

b) Explain. [8]

- i) Importance of planning
- ii) Algorithm for classical planning

c) Explain Limits of AI and Future opportunities with AI. [5]



Total No. of Questions : 8]

SEAT No. :

P-7547

[Total No. of Pages : 3

[6180]-55

T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - II) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer four questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

Q1) a) List All problem solving strategies. What is backtracking, explain with n queen problem. [8]

b) Write Minimax Search Algorithm for two players. How use of alpha and beta cut-offs will improve performance? [9]

OR

Q2) a) Define Game theory, Differentiate between stochastic and partial games with examples. [9]

b) Define is Constraint satisfaction problem, State the types of consistencies Solve the followiig Crypt Arithmetic Problem. [8]

$$\begin{array}{rccccccc} & B & A & S & E & & \\ + & B & A & L & L & & \\ \hline & G & A & M & E & S & \end{array}$$

Q3) a) What is an Agent. Name any 5 agents around you Explain Knowledge based agent with Wumpus World. List and explain in short the various steps of knowledge engineering process [9]

Consider the following axioms: If a triangle is equilateral then it is isosceles.

b) If a triangle is isosceles, then its two sides AB and AC are equal. If AB and AC are equal, then angle B and C are equal. ABC is an equilateral triangle. Represent these facts in predicate logic. [9]

OR

P.T.O.

Q4) a) Write the following sentences in FOL (using types of quantifiers) [9]

- i) All birds fly
- ii) Some boys play cricket
- iii) A first cousin is a child of a parent's sibling
- iv) You can fool all the people some of the time and some of the people all the time, but you cannot fool all the people all the time

b) What is Knowledge Representation using propositional Logic? Compare propositional and predicate Logic. [9]

Q5) a) Explain Forward Chaining and Backward Chaining. With its Properties, advantages and Disadvantages. [9]

Explain :

- b) i) Unification in FOL [8]
- ii) Reasoning with Default information

OR

Q6) a) Explain FOL inference for following Quantifiers [8]

- Universal Generalization
- Universal Instantiation
- Existential Instantiation
- Existential introduction

b) What is Ontological Engineering, in details with its categories object and Model. [9]

Q7) a) Explain with an example Goal Stack Planning (STRIPS algorithm). [5]

- b) Explain with example, how planning is different from problem solving. [5]
- c) Explain AI components and AI architecture. [8]

OR

- Q8)** a) Explain Planning in non deterministic domain. [5]
- b) Explain [5]
- i) Importance of planning.
- ii) Algorithm for classical planning .
- c) What is AI Explain. Scope of AI in all walks of Life also explain Future opportunities with AI. [8]



Total No. of Questions : 8]

SEAT No. :

PB3786

[6262]-45

[Total No. of Pages :2

T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester- II) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer four questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume Suitable data if necessary.

- Q1)** a) List all problem solving strategies. What is backtracking, explain with n queen problem. **[8]**
- b) Write Minimax Search Algorithm for two players. How use of alpha and beta cut-offs will improve performance? **[9]**

OR

- Q2)** a) Define Game theory, Differentiate between stochastic and partial games with examples. **[9]**
- b) Define is Constraint satisfaction problem, state the types of consistencies solve the following Crypt Arithmetic Problem. **[8]**

$$\begin{array}{r} \text{B A S E} \\ + \text{B A L L} \\ \hline \text{G A M E S} \end{array}$$

- Q3)** a) What is an Agent? Name any 5 agents around you explain knowledge based agent with Wumpus World.
List and explain in short the various steps of knowledge engineering process
Consider the following axioms:
If a triangle is equilateral then it is isosceles. **[9]**
- b) If a triangle is isosceles, then its two sides AB and AC are equal.
If AB and AC are equal, then angle B and C are equal.
ABC is an equilateral triangle.
Represent these facts in predicate logic. **[9]**

OR

P.T.O.

- Q4)** a) Write the following sentences in FOL (using types of quantifiers) [9]
 i) All birds fly
 ii) Some boys play cricket
 iii) A first cousin is a child of a parent's sibling
 iv) You can fool all the people some of the time, and some of the people all the time, but you cannot fool all the people all the time.
 b) What is Knowledge Representation using propositional logic?
 Compare propositional and predicate Logic. [9]
- Q5)** a) Explain Forward Chaining and Backward Chaining. With its properties, advantages and disadvantages. [9]
 b) Explain: [8]
 i) Unification in FOL
 ii) Reasoning with Default information
- OR
- Q6)** a) Explain FOL inference for following Quantifiers. [8]
 i) Universal Generalization
 ii) Universal Instantiation
 iii) Existential Instantiation
 iv) Existential introduction
 b) What is Ontological Engineering, in details with its categories object and Model. [9]
- Q7)** a) Explain with an example Goal Stack Planning (STRIPS algorithm). [5]
 b) Explain with example, how planning is different from problem solving. [5]
 c) Explain AI components and AI architecture [8]
- OR
- Q8)** a) Explain Planning in non deterministic domain. [5]
 b) Explain. [5]
 i) Importance of planning
 ii) Algorithm for classical planning
 c) What is AI explain scope of AI in all walks of Life also explain future opportunities with AI. [8]



Total No. of Questions : 5]

SEAT No. :

PC-1728

[Total No. of Pages : 2

[6353] - 45

T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - II) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn whenever necessary
- 3) Assume suitable data, if necessary

- Q1)** a) Define Game Theory. What are the components, defining game as a Search Problem? Draw a Game Tree for tic-tac-toe. [9]
- b) What do you understand by Constraint Propagation? Write short notes on Node Consistency and Arc Consistency. [8]

OR

- Q2)** a) Explain how Minimax and alpha-beta algorithms change for two-players, non zero-sum games in which each player has his or her own utility function. [9]
- b) Define Constraint Satisfaction Problem. Explain Map Coloring Example Problem. . Formulate the Map Coloring Problem as CSP. [8]
- Q3)** a) Define Knowledge base and Sentence. Describe in detail about Wumpus World Environment along with brief description to find out the agent. Explain Task Environment. [8]
- b) Represent the followings into First Order Logic form: [10]
- i) All employees earning Rs.45000 or more pay tax.
 - ii) Sita is a marine engineer and she is also an artist.
 - iii) Children love icecream.
 - iv) If Humidity is high, temperature is high then a person cannot feel comfortable.
 - v) Puppies are cute.
- If AB and AC are equal, then angle B and C are equal.
ABC is an equilateral triangle.
Represent these facts in predicate logic

P.T.O.

OR

- Q4) a)** Write Short notes on followings: [9]
- i) Syntax and Semantics
 - ii) Proposition Logic Vs First Order Logic
 - iii) Knowledge Engineering Process in First Order Logic
- b)** Show the following Sentences are valid or not. [9]
- a) $(P \wedge Q) \rightarrow (P \vee Q)$ b) $(\neg A \vee B) \wedge (\neg B \vee C) \rightarrow (\neg A \vee C)$
- Q5) a)** Prove that Universal Instantiation is sound and that Existential Instantiation produces an inferentially equivalent knowledge base. [8]
- b)** Write Short notes on : [9]
- i) Forward Chaining
 - ii) Categories and Objects
 - iii) Back ward Chaining

OR

- Q6) a)** What do you understand by Resolution? Describe the Procedure to convert into CNF? Find the clause of the expression: $(\neg P \vee Q) \rightarrow R$ [9]
- b)** What are the reasoning systems for Categories? Explain Semantic Network and Description Logic. [8]
- Q7) a)** Describe the differences and similarities between problem solving and planning. [5]
- b)** Explain AI components and AI architecture [5]
- c)** What are the different types of planning? How planning algorithm can be represented as state space search? [8]

OR

- Q8) a)** Explain What are Planning approaches? [6]
- b)** Explain The Blocks World in detail. [6]
- c)** What are the limitations of AI? Explain What are the Future Scopes with AI? [6]



Total No. of Questions : 8]

SEAT No. :

PD4250

[Total No. of Pages : 2

[6403]-45

T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - VI) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Explain Alpha-Beta tree search and cut off procedure with example. [9]
b) Explain in details the concept of backtracking and constraint propagation to solve n-queens problem. [8]

OR

- Q2)** a) What is constraint satisfaction problem? Explain with example. [5]
b) Compare and contrast the stochastics games and partial observable games. [4]
c) How AI technique is used to solve tic-tac-toe problem. [8]

- Q3)** a) Explain Wumpus world environment with PEAS description. [9]
b) What is knowledge representation in propositional logic? Compare and contrast PL and FOL. [8]

OR

- Q4)** a) Explain different inference rules in FOL with suitable example. [9]
b) What is an agent? Explain knowledge based agent with architecture diagram, also state the significance of inference engine. [8]

- Q5)** a) Illustrate with an example the use of the unification algorithm to prove the concept of resolution. [9]
b) Define and explain the forward changing with example, analyze the differences between forward and backward changing. [9]

OR

P.T.O.

Q6) a) Describe Ontological Engineering w.r.t. Categories, Objects and Model. [9]

b) Define First order Logic, Explain FOL inference for following classifiers. [9]

i) Universal Generalization

ii) Universal Instantiation

iii) Extential Instantiation

iv) Extential Introduction

Q7) a) Explain the algorithm for classical planning with an example. [6]

b) Analyze various planning approaches in AI. [6]

c) Explain the Hierarchical planning with relevant example. [6]

OR

Q8) a) Explain with example how planning is different than problem solving. [6]

b) Explain types of planning in details. [6]

c) Explain AI components and AI Architecture. [6]

